D Error Analysis

D.1 Performance Analysis of Premises & Conclusion

We compare the confusion matrix of GPT-4 under the top three best performed prompting schemes and the confusion matrix of GPT-4 under the multi-round prompting scheme of Premises & Conclusion on each dataset. We find that, under the Premises & Conclusion scheme, GPT-4 is significantly biased to over predict "No Fallacy" when this label is given or fallacy types that contain words related to the semantics of "causality", such as"causal" and "generalization". We speculate that the prompt formulation of Premises & Conclusion by offering the formal definition of the term fallacy "*whether the premises entail the conclusion*" has a misleading implication that renders GPT-4 to place excessive attention on fallacy conceptions with semantics that overlap with the word "entail". This is why Premises & Conclusion prompting scheme underperforms the other prompting methods.

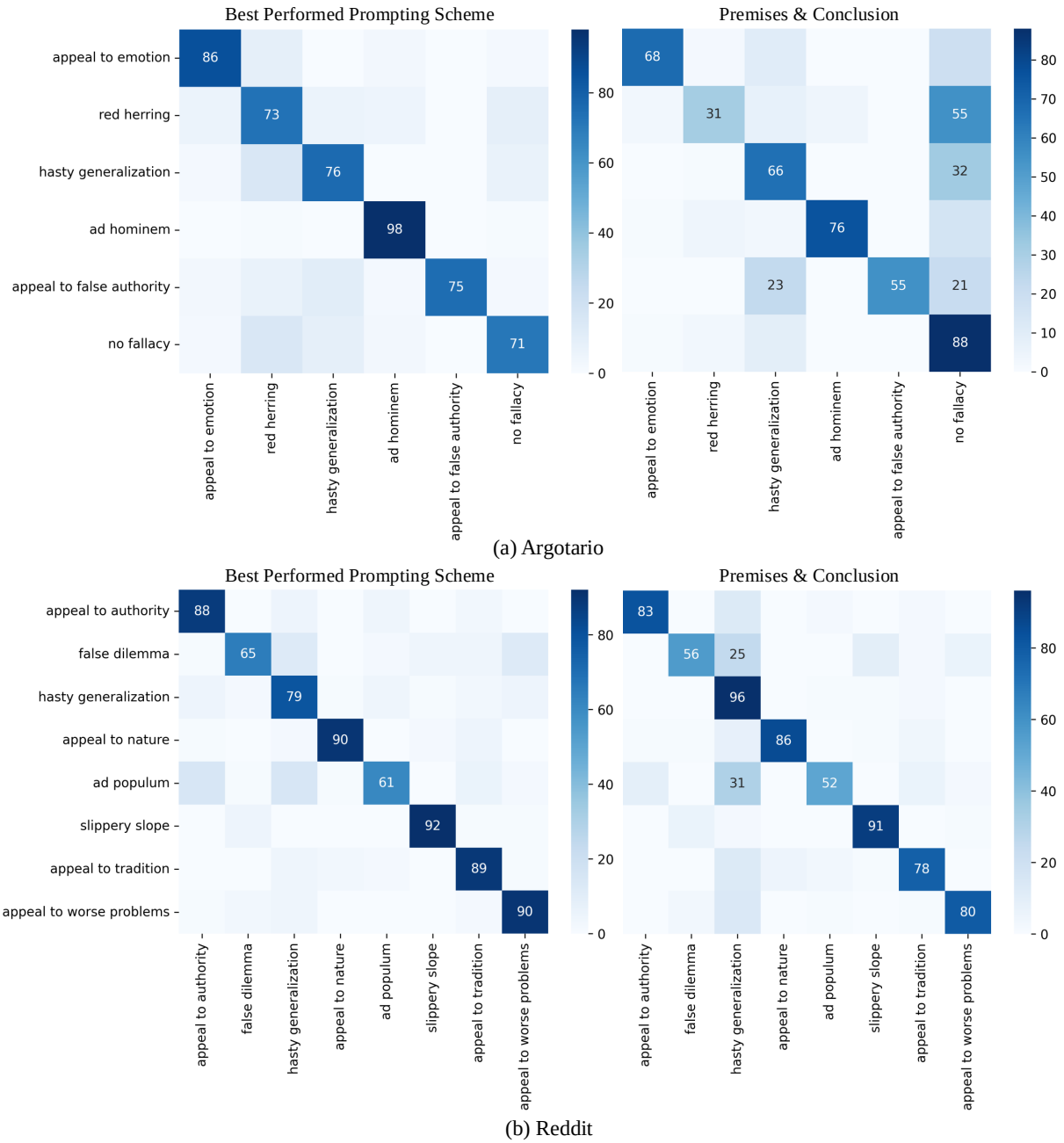
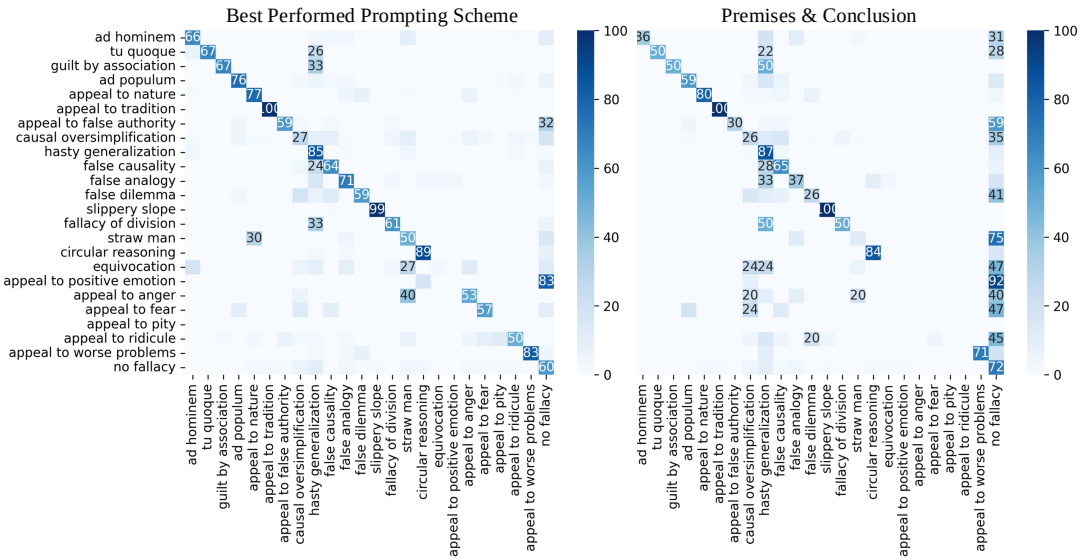
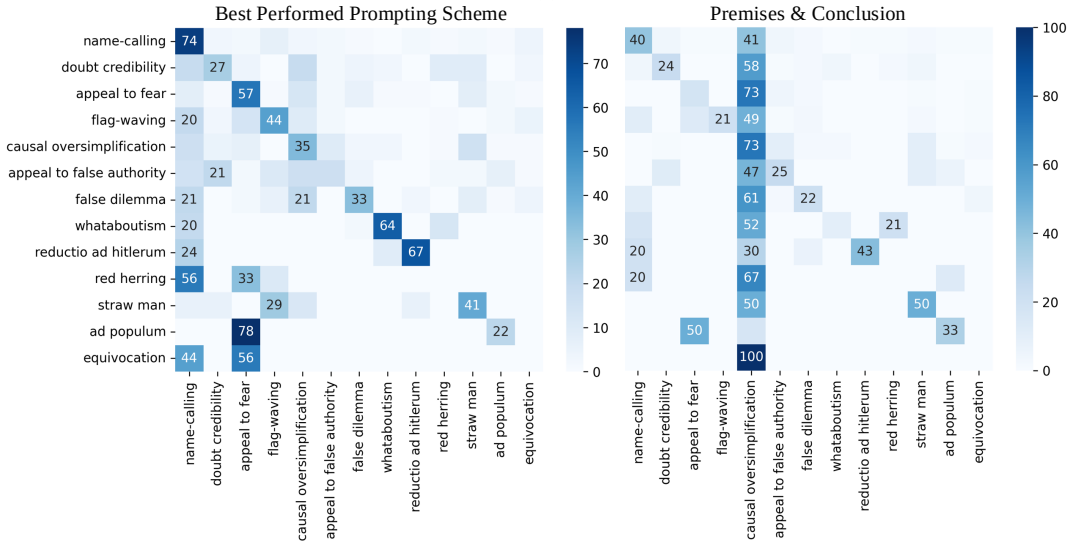


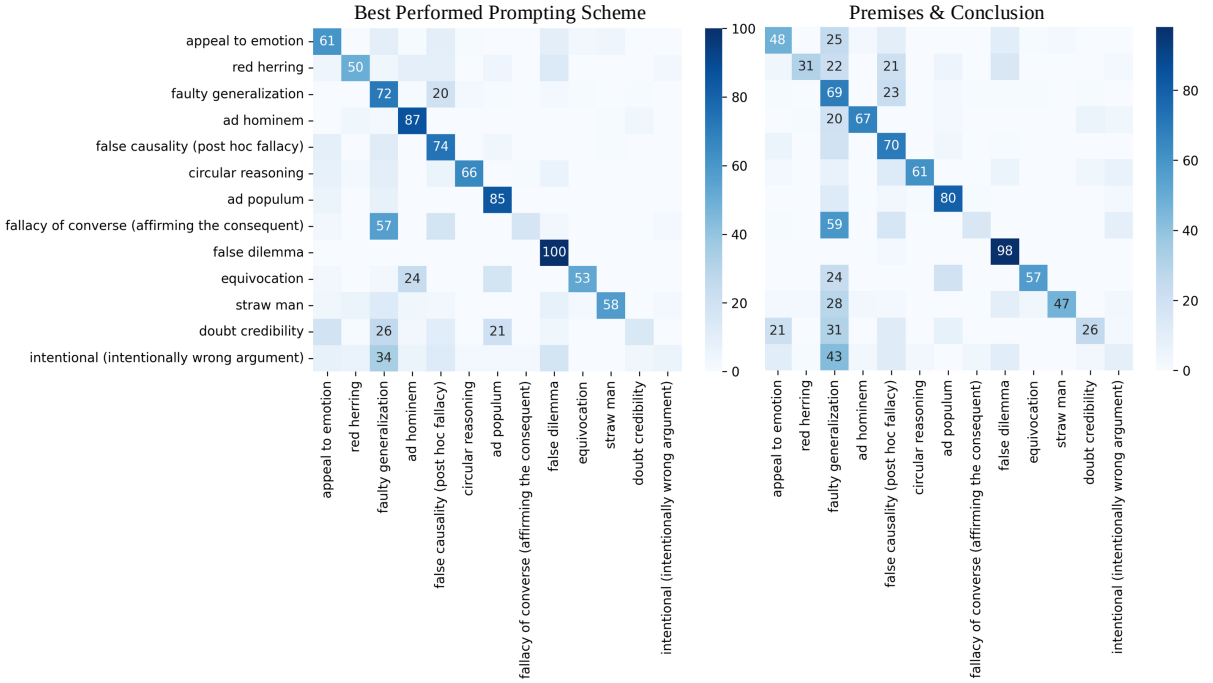
Figure 4: Comparison of GPT-4's results under the best performed prompting scheme and the multi-round Premise & Conclusion on ARGOTARIO and REDDIT.



(c) Mafalda

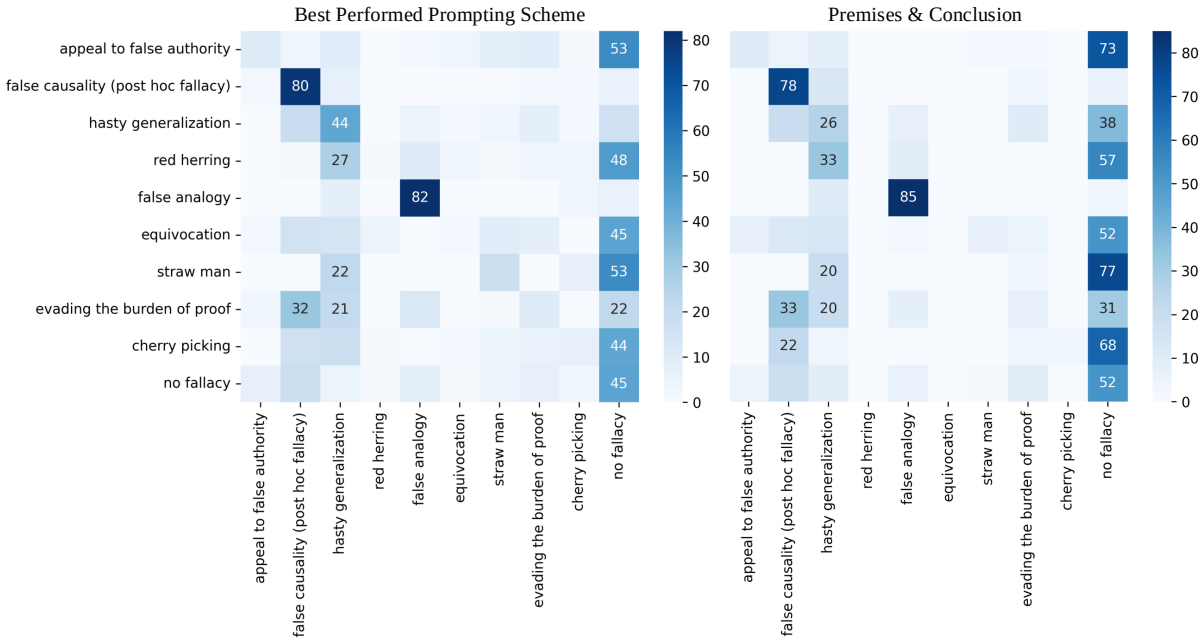


(d) Propaganda

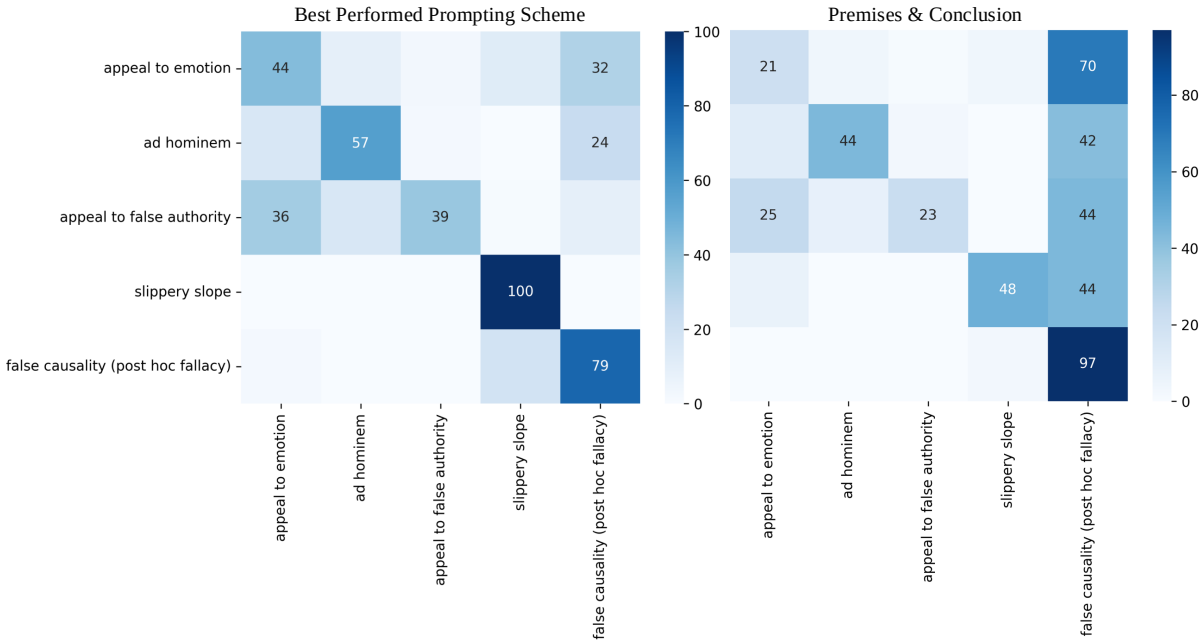


(e) Logic

Figure 5: Comparison of GPT-4's results under the best performed prompting scheme and the multi-round Premise & Conclusion on MAFALDA, PROPAGANDA and LOGIC.



(f) Covid-19



(g) ElecDeb

Figure 6: Comparison of GPT-4's results under the best performed prompting scheme and the multi-round Premise & Conclusion on COVID-19 and ELECDEB.